

Spinal pathways, touch and pain

The written notes for the spinal pathways, touch and pain walkthrough. Week 5 competencies 5, 13 and 14.

Step by step: where the signal travels

3 min

Fine touch, vibration and proprioception: the dorsal column pathway, neuron by neuron

1. First neuron: starts at a receptor in the skin, muscle or joint. Its cell body is in the dorsal root ganglion. Its axon enters the cord through the dorsal root and climbs up the dorsal column on the same side, without synapsing in the cord.
2. First synapse: in the medulla, in the dorsal column nuclei.
3. Second neuron: its axon crosses the midline in the medulla and climbs on the opposite side to the thalamus.
4. Second synapse: in the thalamus.
5. Third (tertiary) neuron: runs from the thalamus to the primary somatic sensory cortex, the postcentral gyrus, on the side opposite the body part. Branches also go to the cerebellum.

Pain, temperature and coarse touch: the spinothalamic tract, neuron by neuron

1. First neuron: starts at a nociceptor or thermoreceptor. Its cell body is in the dorsal root ganglion. Its axon enters through the dorsal root and synapses almost right away in the dorsal horn.
2. First synapse: in the dorsal horn of the spinal cord, within a segment or two of where it entered.
3. Second neuron: its cell body is in the dorsal horn. Its axon crosses the midline inside the cord and climbs on the opposite side all the way to the thalamus.
4. Second synapse: in the thalamus.
5. Third (tertiary) neuron: runs from the thalamus to the somatosensory cortex on the side opposite the body part. Branches also go to the limbic system and hypothalamus.

Voluntary movement: the corticospinal tract, neuron by neuron

1. Upper motor neuron: its cell body is in the primary motor cortex, the precentral gyrus.

2. Its axon runs down through the brainstem to the bottom of the medulla, to the pyramids.
3. Most of the axons, about 85 to 90 percent, cross the midline there.
4. They descend on the opposite side of the spinal cord, in the lateral white matter.
5. Synapse: in the ventral horn, on a lower motor neuron, directly or through an interneuron.
6. Lower motor neuron: its cell body is in the ventral horn. Its axon leaves through the ventral root, runs in the spinal nerve, and ends on the muscle at a neuromuscular junction.

The pain gate, step by step (the classic gate control model)

1. A bumped shin fires nociceptors. Their signal reaches the dorsal horn and excites the projection neuron that carries pain up to the brain.
2. Rubbing the shin fires the large touch fibers from the same skin.
3. The touch fibers excite an inhibitory interneuron in the dorsal horn.
4. The interneuron inhibits the projection neuron.
5. Fewer pain signals leave the dorsal horn for the brain, so the pain feels less.

The body's own painkillers, step by step

1. Descending pathways from the brainstem activate interneurons in the dorsal horn.
2. The interneurons release enkephalin, an endogenous opioid.
3. Before the synapse: enkephalin binds opioid receptors on the nociceptor's terminal, less Ca^{2+} enters, and less substance P and glutamate are released.
4. After the synapse: enkephalin binds opioid receptors on the projection neuron, K^{+} channels open, and the neuron hyperpolarizes.
5. Fewer pain signals travel up the spinothalamic tract.

Referred pain, step by step

1. Heart muscle without enough oxygen fires visceral pain fibers.

2. They enter the same segments of the spinal cord as sensory fibers from the skin of the left arm.
3. Both sets of fibers converge on the same projection neurons in the dorsal horn.
4. The signal climbs to the brain on that shared line.
5. The brain usually receives signals on that line from the skin, so it places the pain in the left arm.

Inside the spinal cord

2 min

- Four regions, named for the vertebrae beside them: cervical, thoracic, lumbar and sacral. Each region has segments, and each segment gives rise to a pair of spinal nerves.
- Just before joining the cord, each spinal nerve splits into a dorsal root, sensory in, whose dorsal root ganglion holds the sensory cell bodies, and a ventral root, out to muscles and glands.
- Gray matter, H-shaped, in the center: cell bodies, dendrites, unmyelinated axons and synapses. A cluster of cell bodies in the CNS is a nucleus; outside the CNS it is a ganglion.
- Dorsal (posterior) horn: sensory fibers synapse on interneurons, in separate somatic and visceral sensory nuclei. Ventral (anterior) horn: motor neuron cell bodies, in somatic motor and autonomic nuclei; the autonomic nuclei sit in the small lateral horn.
- White matter: mostly myelinated axons, bundled into tracts, the CNS equivalent of nerves. A group of tracts is a column.
- Ascending tracts, sensory and going up, run in the dorsal and outer lateral white matter. Descending tracts, mostly motor and going down, run in the ventral and inner lateral white matter. Proprioceptive tracts stay within the cord.
- Most corticospinal axons cross at the pyramids and descend in the lateral corticospinal tract; those that do not cross descend in the anterior corticospinal tract. Because of the pyramids, the corticospinal tract is also called the pyramidal tract.
- Interneurons lie entirely within the CNS. In the cord they route sensory information up to the brain, bring commands down to motor neurons, and modify information as it passes through.
- The cord is a self-contained integrating center for spinal reflexes, which usually also send sensory information up to the brain.
- Damage to a dorsal root ganglion (shingles) affects sensation in one strip of skin. Damage to the ventral horn (polio) causes weakness with sensation spared.

Part	Gray or white matter	What it is
Horn	Gray	A region of cell bodies and synapses
Root	Neither: outside the cord	A branch of a spinal nerve just before it joins the cord
Tract	White	A bundle of axons running up or down
Column	White	A group of tracts

Three long pathways

1 min

Tract	Carries	Direction	Where it crosses the midline	First synapse
Dorsal column	Fine touch, vibration, proprioception	Up	Medulla	Medulla (dorsal column nuclei)
Spinothalamic	Pain, temperature	Up	Spinal cord, within a segment or two of entry	Dorsal horn of the cord
Corticospinal	Voluntary movement	Down	Bottom of the medulla (the pyramids)	Ventral horn motor neuron

- Both sensory pathways relay in the thalamus before reaching the somatosensory cortex.
- Corticospinal neurons in the cortex are upper motor neurons; the motor neurons in the ventral horn are lower motor neurons.
- Each side of the brain senses and moves the opposite side of the body.

Brown-Séquard syndrome: a cut through one half of the cord

- Same side as the cut, below it: loss of fine touch, vibration and position sense, and loss of voluntary movement.

- Opposite side, below it: loss of pain and temperature.
- The split pattern locates the lesion in the cord. Above the medulla all three pathways have crossed, so a brain lesion takes everything from the same side of the body.

Simple reflexes such as the knee jerk are integrated in the spinal cord or the brain stem, but the brain can still modify or override them.

Where these pathways go in the brain 1 min

Region	Role on these pathways
Medulla, at the pyramids	Most corticospinal axons cross the midline here, so each side of the brain controls the opposite side of the body
Pons	Relay between the cerebrum and the cerebellum
Midbrain	Eye movement; relays for visual and auditory reflexes
Reticular formation, through the brain stem	Arousal, muscle tone and stretch reflexes, and pain modulation
Thalamus	Relay and integrating station for almost all sensory information on its way to the cerebrum; smell is the exception
Cerebellum	Receives muscle and joint position information and coordinates movement and posture automatically
Primary somatic sensory cortex, parietal lobe	Touch, temperature, pain, itch and body position. Damage dulls sensation on the opposite side of the body
Primary motor cortex and motor association area, frontal lobe	Start voluntary movement; pyramidal cells send their axons down the corticospinal tract
Visual cortex, occipital lobe	Vision
Auditory cortex, temporal lobe	Hearing
Olfactory cortex, a small region of the temporal lobe	Smell
Gustatory cortex, in the insula	Taste

Somatic senses and their receptors 1 min

Four somatic senses: touch, proprioception, temperature, and nociception (pain and itch). All are transduced by mechanical, thermal or chemical energy opening ion channels, such as Piezo channels for pressure and TRP channels for temperature and pain. Primary sensory neurons are pseudounipolar, with cell bodies in the dorsal root ganglia.

Receptor	Stimulus	Where	Adaptation
Free nerve endings	Temperature, noxious stimuli, hair movement	Around hair roots and under the skin surface	Variable
Meissner corpuscle	Flutter, stroking	Superficial skin	Rapid
Pacinian corpuscle	Vibration; large receptive fields	Deep skin	Rapid
Ruffini corpuscle	Skin stretch	Deep skin	Slow
Merkel receptor	Steady pressure, texture; Merkel cell with Piezo2 releases transmitter onto the nerve ending; densest in fingertips	Superficial skin	Slow

- Thermoreceptors in the skin are free nerve endings. Cold receptors respond below body temperature; warm receptors from 37 °C (98.6 °F) to about 45 °C (113 °F); above that, nociceptors produce painful heat. Cold receptors outnumber warm ones.
- TRPV1 channels open with damaging heat and with capsaicin (chili peppers). TRPM8 opens with cold and with menthol.
- The somatosensory cortex, in the parietal lobe, has a sensory field for each body part, with columns for each kind of receptor. Area tracks sensitivity, and the map changes with use (Braille readers) and after loss of a body part (phantom limb pain when

reorganization goes wrong). The brain has no pain receptors.

Pain and its modulation

2 min

Fiber	Structure	Speed	Carries
A-beta	Large, myelinated	30 to 70 m/s	Mechanical stimuli (touch)
A-delta	Small, myelinated	12 to 30 m/s	Fast pain, cold, mechanical stimuli
C	Small, unmyelinated	0.5 to 2 m/s	Slow pain, heat, cold, mechanical stimuli, itch

- Nociceptors: free nerve endings in skin, joints, muscles, bones and organs, not in the brain or spinal cord. They respond to stimuli that cause or could cause tissue damage.
- Fast pain is sharp and localized (A-delta); slow pain is dull and diffuse (C). Itch (pruritus) comes from a subtype of nociceptor fiber; scratching seems to interrupt it.
- Nociceptor fibers end in the dorsal horn and drive two things: protective spinal reflexes such as withdrawal, and ascending pathways to the thalamus and cortex. Branches to the limbic system and hypothalamus add suffering and autonomic reactions (nausea, sweating, fainting).
- Deep somatic pain comes from muscle, as in the burn of exercise or ischemia. Visceral pain is poorly

localized.

- Referred pain: visceral and somatic nociceptors converge on one ascending tract; the brain assigns the pain to the skin. Heart: neck, shoulder, left arm. Gallbladder irritating the diaphragm: right shoulder and neck.
- Inflammatory pain: histamine, prostaglandins and substance P at an injury activate or sensitize nociceptors by lowering their threshold. Aspirin inhibits prostaglandins.
- Chronic pain lasts weeks or months and reflects changes in the nervous system; neuropathic pain, such as diabetic neuropathy, comes from damage to the somatosensory system.
- Gate control theory: touch fibers excite an inhibitory interneuron that inhibits the pain projection neuron in the dorsal horn. Your textbook notes this classic model rests on inaccurate assumptions and leaves out spinal glia; what holds is that pain can be modulated in the cord.
- Descending pathways from the brain inhibit nociceptor neurons in the cord, as in emergencies.

Endogenous opioids

Endorphins, enkephalins and dynorphins.

Enkephalins and dynorphins are released by neurons in pain pathways; beta-endorphin comes from the anterior pituitary, from the same prohormone as ACTH. Opioid receptors decrease transmitter release from primary sensory neurons and inhibit secondary neurons. Long-term opioid drugs cause tolerance.

Other pain relief

Capsaicin patches (TRP channels), ziconotide (blocks Ca^{2+} channels, given into the CSF), neuromodulation with electrodes, and acupuncture.